University of Engineering and Technology, Lahore

BS Computer Science

Final Year Project

Revised Proposal

Offline Intelligent Support System for Psychological Documentation and Longitudinal Progress Monitoring

With Exploratory Speech-Trend Monitoring
for Bipolar Disorder

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Revised following project research, earlier professional feedback, and consultation with
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1 Executive Summary

This project proposes an offline assistant that helps psychologists prepare structured session notes and monitor a client's progress across multiple sessions. With consent, the system will process session audio locally, generate an editable draft, and present changes in symptoms, everyday functioning, interventions, and client responses.

Following consultation with Ms. Kinza Nazir, the project places greater emphasis on longitudinal monitoring, a practical clinical documentation structure, and explicit psychologist review. Bipolar disorder is the first clinical use case, with speech-rate and pause trends treated as an exploratory supporting feature. Anxiety-related monitoring will initially focus on generalized anxiety and social anxiety.

The system will distinguish client statements, clinician observations, and machine-generated suggestions. Visual observations and clinical judgments will be entered or confirmed by the psychologist. Risk-related content will appear prominently and require explicit review.

Development will begin on the team's existing laptop. A later phase will target the NVIDIA Jetson Orin Nano Super 8GB, subject to measured performance and budget. The project will deliver an evaluated academic prototype; clinical effectiveness will not be inferred from technical performance alone.

Central objective: Reduce the effort of preparing and reviewing session records while helping a psychologist follow changes over time.
Primary outputs: Editable notes, reviewed progress records, source-linked observations, and exploratory speech-trend displays.

2 Problem Statement

A psychological session contains information that must be organized into a useful clinical record: the client's concerns, the clinician's observations, assessment findings, interventions, homework, client responses, and follow-up plans.

In the consultation, Ms. Nazir described documentation as a challenging part of practice and emphasized the usefulness of monitoring changes across sessions. These changes may include symptom frequency, intensity, duration, sleep, appetite, academic performance, and social functioning.

Electronic records can store this information, but the team must still establish which preparation and comparison tasks remain manual in the relevant workflow. The proposed system addresses this specific documentation and review problem.

Local processing is a design choice intended to reduce dependence on external processing services and internet availability. Its practical value and usability will be evaluated alongside note quality and review effort.

3 Clinical Consultation and Scope Revision

3.1 Evidence Used

This revision draws on:

- The previous project proposal and its earlier professional consultation responses.
- The team's direct recollection of the meeting with Ms. Kinza Nazir on 5 October 2026.

- The supplied Gemini-generated meeting summary.

The previous proposal contains inconsistencies in the stated number of earlier respondents. Their qualitative feedback is retained, but respondent percentages and vote-based charts are omitted.

The meeting summary is an automatically generated record, not a verbatim verified transcript. Specific terminology and the final note template will be confirmed with Ms. Nazir. In particular, the electronic note-system name recalled as "ikennotes" has not been verified.

3.2 Consultation Findings and Design Decisions

Consultation finding	**Resulting design decision**
Monitoring changes across sessions was strongly supported.	Make longitudinal symptom and functioning records a core feature.
Documentation includes subjective reports, objective observations, assessment, interventions, responses, and progress.	Use one configurable clinical note template, confirmed with the consultant, rather than implement several competing formats initially.
Speech characteristics may offer supporting information for bipolar monitoring, but their wider usefulness is limited.	Prioritize speech rate and pauses as exploratory trends; avoid automatic mood or emotion labels.
Risk of harm to self or others requires particular attention and clinician approval.	Provide a prominent risk-review section, supporting excerpts, and a mandatory review state before note approval.
Clients may communicate unusually or provide difficult-to-follow accounts.	Preserve source excerpts, mark uncertainty, and avoid rewriting unclear content into an invented coherent account.
Sessions commonly last 45–60 minutes.	Evaluate memory use, processing time, and reliability using recordings of realistic session length.
In-person sessions provide additional observational information; online work introduces privacy and connectivity considerations.	Allow clinician-entered observations in both settings. Keep offline processing distinct from the internet connection used for an online consultation.
Bipolar disorder and anxiety were emphasized; bulimia was not recommended for the present scope.	Prioritize bipolar monitoring and a limited anxiety workflow; exclude a dedicated bulimia module.
Ms. Nazir expressed willingness to support supervision after the expected role is clarified.	Propose clinical co-supervision, subject to university approval and agreement on responsibilities and time commitment.

3.3 Meeting Photograph

4 Narrowed Project Scope

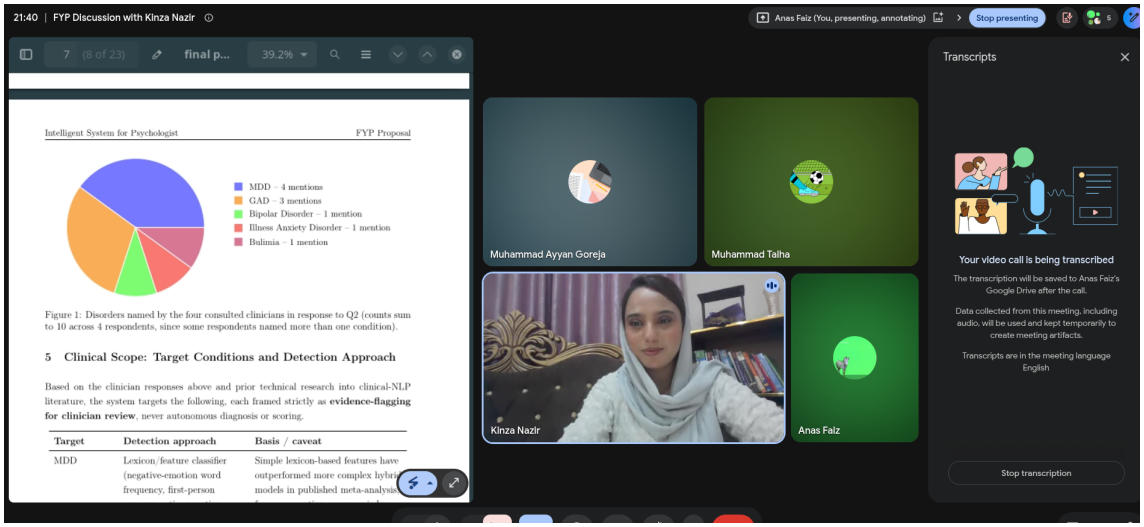


Figure 1: Project consultation with Ms. Kinza Nazir, 5 October 2026. Photograph to be included with permission.

Priority	**Use case**	**Planned support**
Primary	Bipolar disorder monitoring	Reviewed symptom and functioning records, with exploratory within-person speech trends.
Secondary	Generalized anxiety and social anxiety	Tracking reported symptoms, situations, daily impact, interventions, homework, and client responses.
Deferred	Separate depression, illness-anxiety, and other disorder modules	General symptoms may be documented, but dedicated disorder-specific models are outside the first version.

Table 2: Project priorities, not estimates of clinical prevalence or effectiveness.

4.1 Clinical Priorities

Bipolar monitoring refers to supporting a psychologist who already knows the clinical context. The system will not identify bipolar disorder from a recording or independently classify a manic or depressive episode.

The proposed academic evaluation will focus on consenting adult volunteers performing fictional scenarios. Unusual speech, disorganized accounts, and poor recording quality will be included as technical challenge cases where feasible.

4.2 Core Deliverables

1. Consent-controlled audio capture or local audio import.
2. Timestamped transcription with speaker attribution.
3. Editable, source-linked clinical note drafts.
4. Clinician-entered objective observations and assessment fields.
5. Explicit review of risk-related content.
6. Longitudinal symptom and functioning dashboards.
7. Local encrypted storage and controlled export.
8. Performance and quality evaluation on the laptop.

4.3 Exploratory Work

- Speech-rate and pause trends for the bipolar monitoring use case.
- Urdu–English mixed-language transcription evaluation.
- Porting the working pipeline to the Jetson device.

4.4 Outside the Current Scope

Autonomous diagnosis, automatic treatment recommendations, automated crisis management, facial emotion recognition, camera-based mental-status assessment, continuous surveillance between sessions, and integration with hospital billing or electronic medical-record systems are excluded.

Fine-tuning a language model is optional and will only be considered if the baseline approach has a clearly identified limitation and suitable training data are available.

5 Project Objectives

1. Build a complete local workflow from recording to an approved session note.
2. Separate client-reported information, clinician observations, and generated draft content.
3. Preserve the supporting transcript and timestamps for generated statements.
4. Compare reviewed symptom and functioning records over time.
5. Make risk-review status visible and require clinician sign-off.

6. Measure transcription quality, note accuracy, review effort, processing latency, and memory usage.
7. Evaluate whether a compact Jetson deployment can support the selected workload.

6 Proposed Clinical Documentation Structure

The meeting discussed SOAP-style documentation and a more detailed electronic workflow. The following is a proposed project template, not a claim that the consultant uses one particular standardized format. Its final fields will be agreed with her.

Section	Content	Source / responsibility
Session details	Date, duration, session mode, consent status, and reviewed goals.	Entered or confirmed by clinician.
Subjective report	What the client says about symptoms, concerns, experiences, and changes since the previous session.	Drafted from attributed transcript excerpts; reviewed by clinician.
Objective observations	Observed behaviour, appearance, mental status, and coherence of communication.	Primarily clinician-entered. Audio cannot establish appearance, hygiene, gait, or body language.
Assessment	Clinician interpretation, psychological tools used, explicitly collected scores, and symptom severity.	Clinician-entered or confirmed; no autonomous diagnostic conclusion.
Progress and functioning	Changes in frequency, intensity, duration, sleep, appetite, academics or work, and social interaction.	Reviewed client reports and clinician observations.
Interventions and homework	Interventions delivered, motivational work, assigned activities, and agreed actions.	Transcript-supported draft with clinician correction.
Client response	Reported or observed response, engagement, motivation, and difficulties with tasks.	Client reports and clinician observations kept distinct.
Risk review	Relevant statements concerning harm to self or others, clinical assessment, and the clinician's documented response.	Explicit clinician review required.
Plan and approval	Next-session priorities, follow-up actions, amendments, reviewer identity, and approval time.	Clinician-controlled.

Missing information will remain "not documented" rather than being filled with an inferred normal finding. Terms such as "word salad" or other mental-status judgments will only appear when entered or confirmed by the clinician.

7 Risk-Related Documentation and Human Review

RISK REVIEW REQUIRED
Statements potentially concerning harm to self or others will be presented with their source, speaker, and timestamp.
The system will not infer that a client is safe because no statement was flagged. A blank or unreviewed field will remain visibly **not assessed / pending review**.

The proposed review states are:

1. **Pending:** risk-related documentation has not been reviewed.
2. **Review required:** a candidate excerpt or clinician-entered concern needs attention.
3. **Reviewed:** the psychologist has recorded their assessment and any relevant action.

A risk-review state is a workflow status, not a predicted risk level. Red text will be accompanied by a clear label so colour is not the only signal.

Draft notes cannot become approved records until the clinician completes the required review fields. The system does not replace the clinician's assessment during a session and is not an emergency alert service.

8 Longitudinal Progress Monitoring

The dashboard will organize reviewed observations by session and allow the psychologist to inspect the underlying evidence.

- **Symptoms:** frequency, duration, intensity, context, and reported impact.
- **Functioning:** sleep, appetite, academics or work, social interactions, and daily activities.
- **Treatment process:** interventions, homework, participation, and client response.
- **Assessment tools:** explicitly administered measures, with their names, dates, and confirmed scores.

For example, a change from symptoms reported throughout the day to symptoms reported mainly at night would be displayed as a change in reported timing. The psychologist would interpret its significance alongside severity and functioning.

The system will preserve units, question wording, and reporting periods. Missing observations will appear as missing values, not zero severity. Comparisons between different scales will not be presented as equivalent.

9 Exploratory Speech-Trend Analysis

Based on the consultation, acoustic analysis is a supporting research component rather than the central clinical claim.

The initial measurements will include:

- Estimated speech rate using transcript timestamps.
- Pause duration and the proportion of speaking time.

- Pitch variation where audio quality permits reliable extraction.

Comparisons will be within the same speaker across sessions. Microphone position, recording gain, background noise, language, and transcription errors will be recorded as potential confounding factors.

The dashboard will display measurements and recording-quality information. It will not automatically label a person as angry, sad, manic, or depressed.

Jitter and shimmer are deferred until the team can justify their measurement reliability and usefulness for the selected recording conditions.

Role-play recordings can evaluate feature extraction and interface behaviour. They cannot establish whether those features accurately measure clinical mood changes.

10 System Architecture

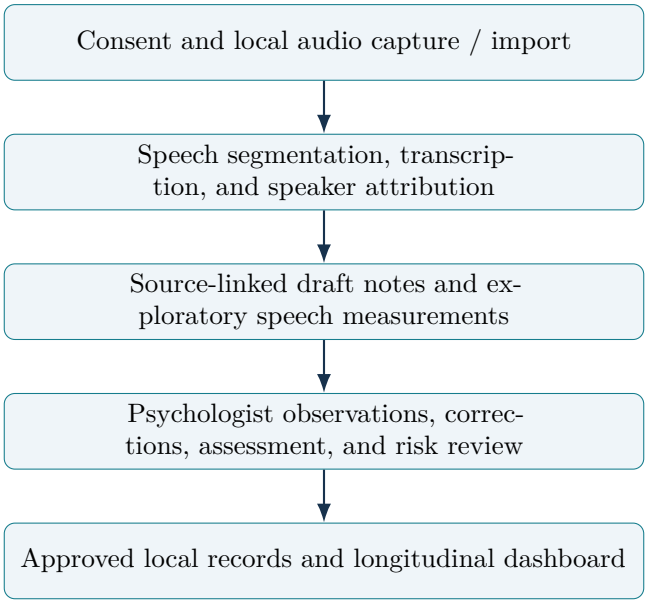


Figure 2: Proposed processing and review workflow. This diagram represents the design, not measured results.

10.1 Processing Strategy

Audio capture must remain reliable even when inference is slower than the conversation. The baseline will therefore support complete recording followed by transcription and note generation.

Chunked transcription during the session may be enabled after benchmarking. Heavy models will run sequentially where necessary. The prototype will not promise instant notes or a fixed completion time before measurements are available.

Candidate speech-recognition models include the multilingual Whisper family [3]. A small quantized language model will be selected through task-specific tests rather than parameter count alone.

10.2 Speaker Attribution

For in-person recordings, two microphones connected to one two-channel USB audio interface are proposed. This provides synchronized channels and may improve speaker separation.

Both microphones can still capture both speakers. Channel identity will therefore not be treated as perfect speaker ground truth. Manual annotations will be used for evaluation, particularly for overlapping speech and cross-talk.

10.3 Evidence and History

Each draft statement will retain its supporting transcript segment where available. The initial history view will retrieve approved structured records by client and session date.

Semantic retrieval over earlier notes may be added after the core workflow works reliably. Unapproved drafts will remain distinct from approved history.

11 Narrowed Hardware Plan

11.1 Phase One: Existing Laptop

The first implementation will use the available HP Victus laptop:

- Intel Core i5-12450H processor.
- 16GB system RAM.
- NVIDIA GTX 1650 with 4GB dedicated GPU memory.
- Existing SSD storage, with free capacity checked before testing.

This phase avoids immediate board procurement and supports rapid development. The separate RAM and GPU-memory limits will be measured; models may require CPU execution or partial GPU offloading.

11.2 Phase Two: Jetson Deployment

The selected compact deployment target is the **NVIDIA Jetson Orin Nano Super 8GB Developer Kit**. It provides shared CPU/GPU memory and supports NVMe storage [4].

Component	Planned configuration
Compute	Jetson Orin Nano Super 8GB Developer Kit.
Storage	256GB NVMe SSD, with capacity and retention reviewed after testing.
Audio input	Two wired microphones suitable for speech capture.
Audio interface	Linux-compatible USB interface exposing two independent input channels; microphone power and connector compatibility verified.
Power and enclosure	Suitable regulated supply, active cooling, and a protective case.
User interface	Local display and keyboard, or a locally attached workstation during development.

Table 4: Planned standalone hardware configuration.

The target is the **Orin Nano Super**, not the older Jetson Nano. Raspberry Pi and Orange Pi alternatives are removed from the primary implementation plan.

11.3 Migration Decision

Jetson procurement and migration will follow:

1. A working laptop demonstration.
2. Measurements on a 45–60 minute recording.
3. Selection of models that meet the intended quality requirements.
4. A memory and runtime estimate for the Jetson.
5. A budget review covering the complete device and audio setup.

If the selected workload cannot meet acceptable performance on the Jetson, the laptop prototype remains the evaluated deliverable, and deployment limitations will be documented.

"Offline" means local processing without internet dependence. Battery-powered operation away from mains electricity is not a current requirement.

12 Consent, Privacy, and Data Handling

- Recording requires documented consent, with visible controls to pause or stop.
- Models will be downloaded during setup; session inference will run locally.
- Local records will use encrypted storage with documented key management and access controls.
- Temporary files, backups, logs, and exports will be included in the data-handling design.
- Raw-audio retention will be configurable and agreed before evaluation.
- Exports will require an explicit user action.
- Access, edits, approvals, exports, and deletion actions will be logged locally.

For an online consultation, the call platform itself uses the network. Only the proposed assistant's processing is local. Privacy testing will therefore distinguish an isolated offline workflow from operation alongside an online call.

The initial evaluation will use fictional scenarios and consenting volunteers. Required university ethics procedures will be checked before recording. Any later use of real clinical records would require separate approval and an appropriate data-governance plan.

13 Evaluation Plan

No benchmark results are claimed in this proposal. Numerical acceptance thresholds will be agreed after a baseline pilot and before the final evaluation.

Dimension	Planned measurement
Transcription	Word error rate against manually checked transcripts; English and Urdu–English results reported separately, with documented text normalization.
Speaker attribution	Speaker-label errors, including cross-talk and overlapping speech, against manual annotations.

Dimension	Planned measurement
Draft accuracy	Unsupported statements, omissions, incorrect attribution, and agreement with reference notes.
Clinical usefulness	Consultant review of structure, relevance, clarity, and required corrections.
Documentation effort	Time to prepare a note manually compared with time to review and correct a generated draft on comparable scenarios.
Risk-related extraction	Precision, recall, and error examples on scripted positive, negative, negated, and historical statements; mandatory review workflow tested separately.
Progress tracking	Correct storage and comparison of dates, units, symptom descriptions, scores, and missing values.
Speech measurements	Repeatability and agreement with manually checked timing measures; sensitivity to recording conditions.
Hardware	Peak RAM/VRAM use, processing time, capture stability, storage use, and sustained-load behaviour.
Privacy	Check for unintended outbound traffic in the defined test workflow, alongside access, export, and deletion checks.

Scenarios used to tune prompts will be separated from final test scenarios. Dataset size, language mix, reviewer count, and uncertainty will be reported. A small role-play study will support claims about prototype behaviour, not diagnostic accuracy or treatment outcomes.

14 Technical Contribution and FYP Value

The contribution is the design and evaluation of a locally operating clinical documentation workflow that combines:

1. Audio capture and timestamped speaker-attributed transcription.
2. Evidence-linked generation with uncertainty preserved.
3. Separation of client reports and clinician observations.
4. Mandatory review and traceable approval.
5. Structured longitudinal symptom and functioning records.
6. Measured deployment trade-offs between a laptop and compact edge hardware.

The project does not claim that local transcription or AI-assisted notes are themselves new. The literature review will compare the implemented workflow with documented capabilities of relevant systems. Earlier blanket claims that competing tools lack all local processing or longitudinal features are not retained.

The research questions are:

- Can source-linked drafts reduce documentation effort while preserving clinically relevant information?

- Can reviewed records make session-to-session changes easier to inspect?
- Which quality and runtime trade-offs arise when the pipeline moves to compact hardware?
- Are speech-trend displays understandable and useful enough to justify further clinical research?

15 Proposed Nine-Month Work Plan

The schedule is relative to approval of the revised scope and will be aligned with departmental submission dates.

Month	Work and milestone
1	Confirm clinical template, scope, supervision role, data plan, and evaluation protocol. Complete requirements specification.
2	Build consent controls, local recording/import, storage foundation, and transcription baseline on the laptop.
3	Add speaker attribution, timestamps, correction tools, and audio quality checks. Demonstrate recording-to-transcript workflow.
4	Implement structured draft notes, supporting excerpts, manual objective fields, and clinician approval.
5	Implement symptom/functioning history, progress displays, intervention records, and explicit risk-review workflow.
6	Evaluate Urdu–English transcription and exploratory speech measurements. Conduct consultant review of fictional outputs.
7	Optimize memory and latency; attempt Jetson migration after the hardware decision checkpoint.
8	Run the held-out technical and usability evaluation. Resolve major correctness and privacy issues.
9	Finalize the report, deployment instructions, user guide, limitations, and demonstration.

16 Feasibility and Main Project Risks

Risk	Response
Inaccurate transcription	Evaluate realistic language and recording conditions; retain correction controls and supporting audio.
Invented or omitted note content	Use source-linked drafts, explicit missing fields, and clinician review; measure both errors and omissions.
Misleading speech trends	Keep the feature exploratory, document confounders, and avoid automatic mood-state labels.
Limited device memory	Run heavy stages sequentially, limit context size, and benchmark before hardware procurement.
Too much clinical scope	Prioritize the shared documentation workflow, bipolar monitoring, and limited anxiety tracking.
Limited reviewer availability	Agree small review batches early and document the amount of clinical input actually obtained.

The project is technically plausible as an academic prototype. Its feasibility depends on a manageable scope, measured model performance, and regular review of the clinical template.

17 Clinical Co-Supervision and Recognition

Ms. Kinza Nazir expressed willingness to support the project, subject to clarification of expectations. Formal appointment as an external or industry co-supervisor remains subject to university requirements and mutual agreement.

The proposed clinical role includes:

- Reviewing the note structure and monitoring fields.
- Advising on interpretation limits and risk-review presentation.
- Reviewing a small agreed sample of fictional session outputs.
- Providing feedback on dashboard usability and evaluation criteria.

The team will provide a written summary of expected involvement, meeting frequency, review workload, and university documentation.

Her contribution will be acknowledged, with her consent, in the report and presentation. Any formal co-supervisor listing will follow university approval. Publication authorship, if applicable, will be discussed separately according to actual contribution.

18 Final Deliverables

1. Working offline laptop prototype.
2. Jetson deployment and benchmark report if migration succeeds; otherwise a documented feasibility assessment.
3. Configurable clinical note template and approval workflow.
4. Longitudinal symptom and functioning dashboard.

5. Exploratory speech-trend module with documented limitations.
6. Evaluation dataset description, protocol, and results.
7. Requirements, architecture, source code, and deployment guide.
8. Consultation record, final report, and demonstration.

19 Approval Requested

The team seeks approval of the revised clinical scope, laptop-first development plan, conditional Jetson migration, and proposed external clinical supervision arrangement.

The immediate next step is to confirm the note template and review requirements with Ms. Nazir, then implement and benchmark the recording-to-draft workflow.

References

[1] FYP team and Kinza Nazir. *Clinical consultation on documentation, monitoring, consent, and project supervision.* 5 October 2026. Team recollection and supplied Gemini-generated meeting notes; terminology subject to participant confirmation.

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[4] NVIDIA. *Jetson Orin Nano Super Developer Kit.* Official hardware specifications. https://www.nvidia.com/en-us/autonomous-machines/embedded-systems/jetson-orin/nano-super-developer-kit/